CPEN 416

Lecture 14: QPE and order finding

Thursday 29 October 2026

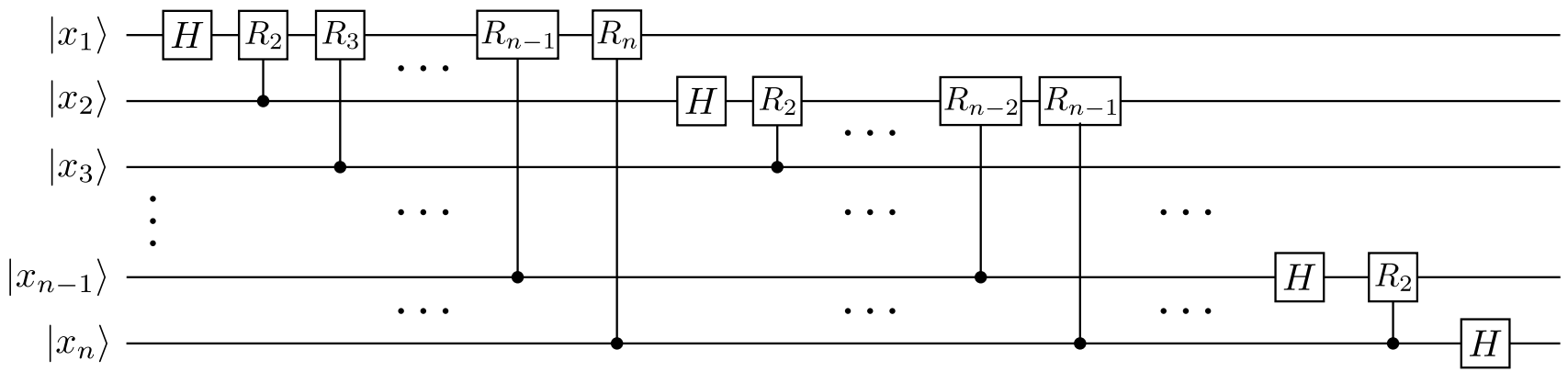
Announcements

- Quiz 7 on Tuesday
- Tutorial Assignment 2 due Friday 23:59
- Midterm checkpoint report due Monday **9:00am**
  - Calendar bookings for checkpoints meetings available
- Practice problem set 3 available (about QPE)

Last time

We implemented the quantum Fourier transform using a circuit with a *polynomial* number of gates and depth.

$$QFT|x\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{xk} |k\rangle$$



Reminder: where are we going?

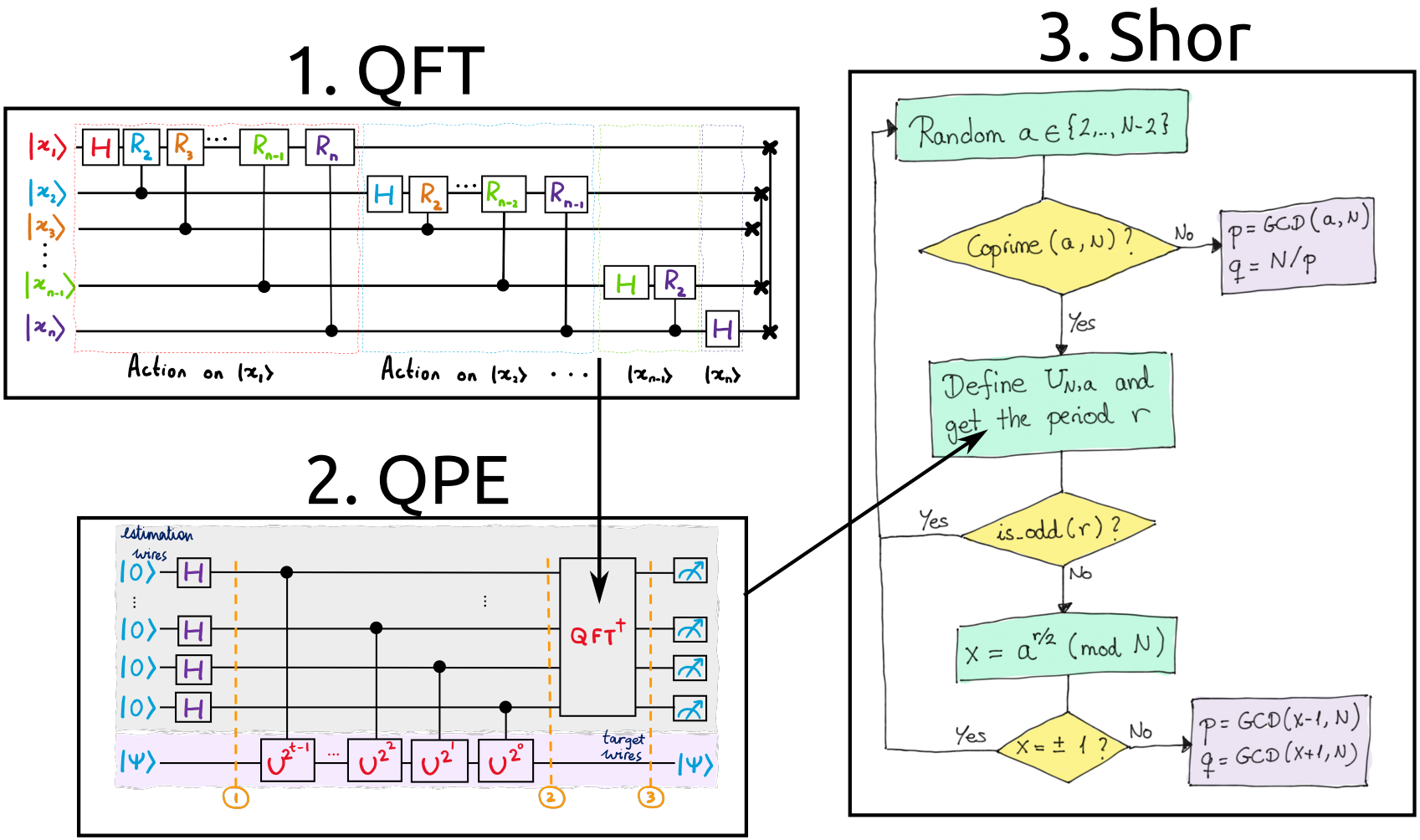


Image credit: Xanadu Quantum Codebook nodes F.3, P.3, S.4

Learning outcomes

- Implement the quantum phase estimation (QPE) subroutine
- Perform arithmetic modulo N with integers and with quantum operations
- Define the order finding problem
- Apply QPE to solve the order finding problem

Eigenvalues of unitary matrices

Eigenvalues of unitary matrices are complex numbers with magnitude 1, i.e.,

where θ_k is a phase, $|\theta_k| \leq 1$. (See proof at end of slides)

What if we want to *estimate* an unknown θ_k?

Quantum phase estimation

Given unitary U and eigenvector $|k\rangle$, estimate θ_k such that

$$U|k\rangle = e^{2\pi i\theta_k}|k\rangle$$

Must determine:

- How to design a circuit that extracts θ_k
- What precision can we estimate it to
- What to do if we don't know $|k\rangle$ in advance

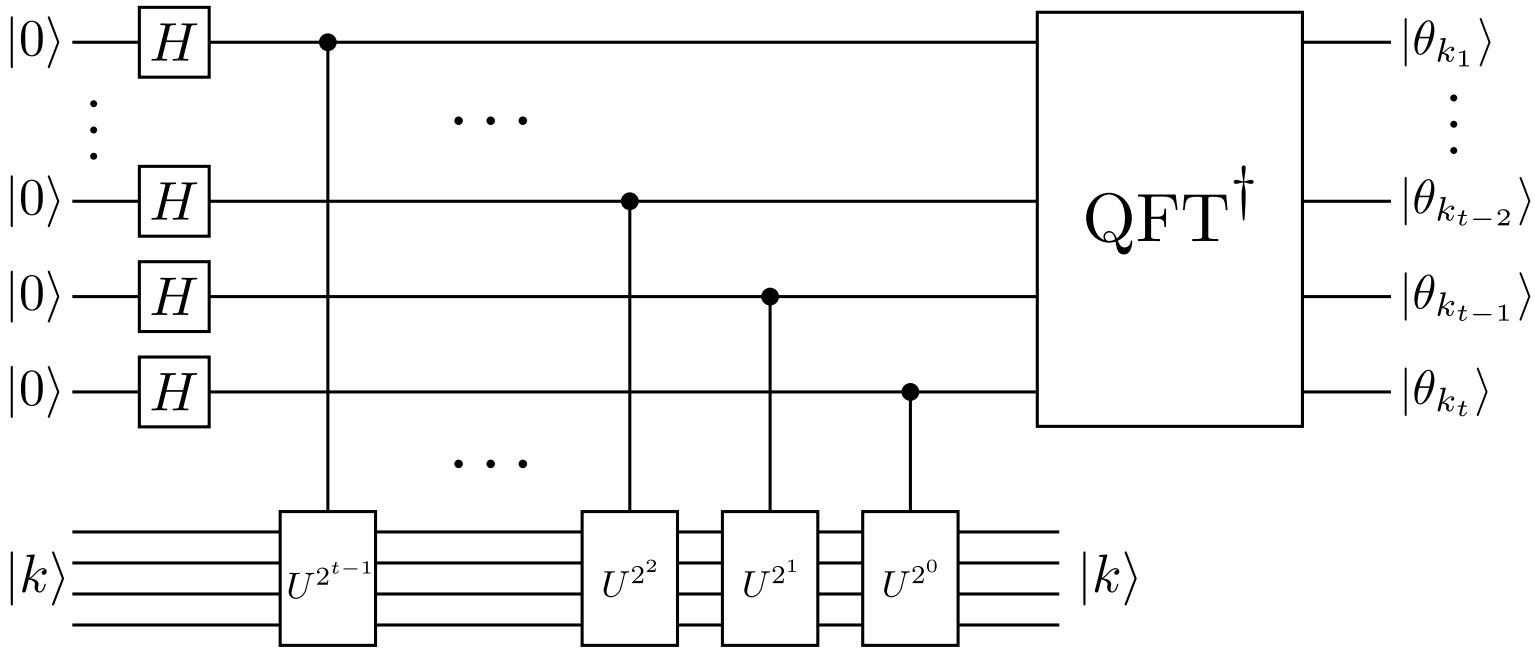
(*You can explore the last two in practice assignment 3!*)

Quantum phase estimation

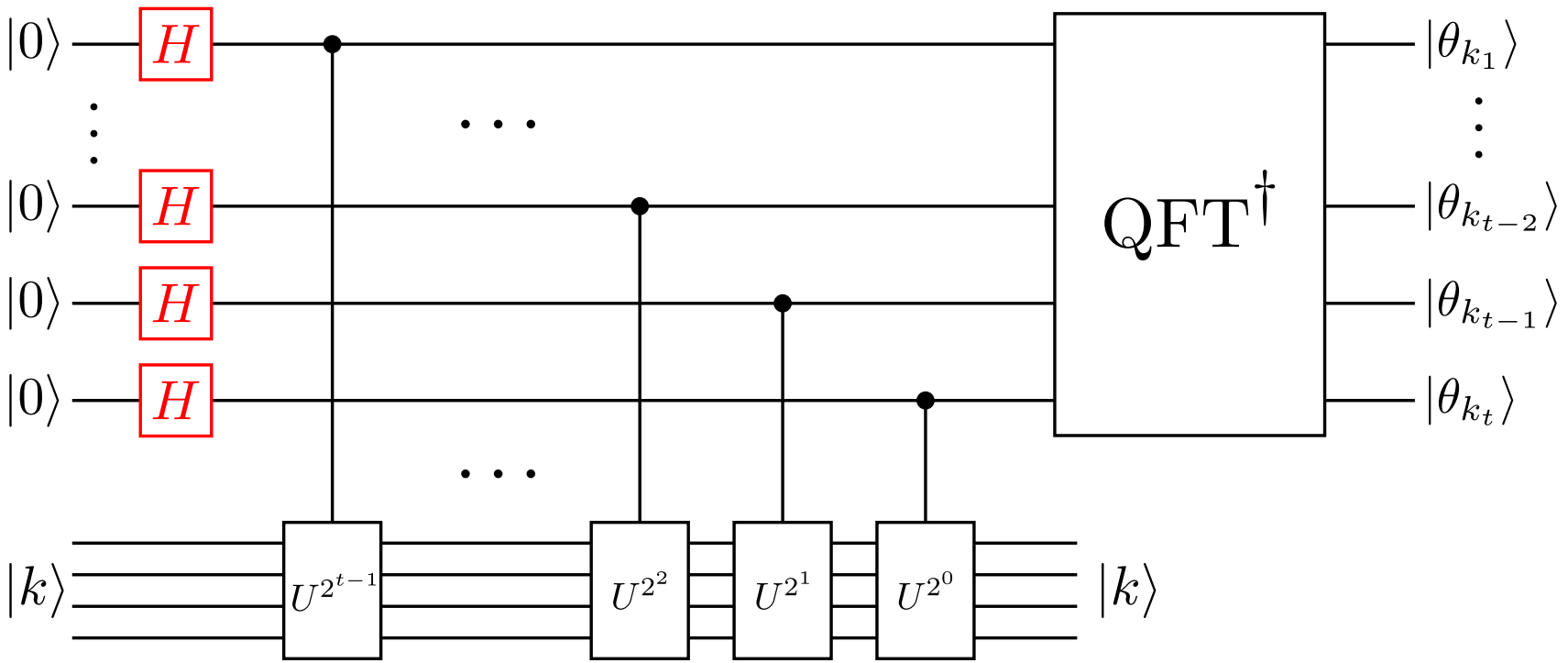
Let U be an n-qubit unitary; $|k\rangle$ is an n-qubit eigenstate.

Assume θ_k can be represented *exactly* using t bits:

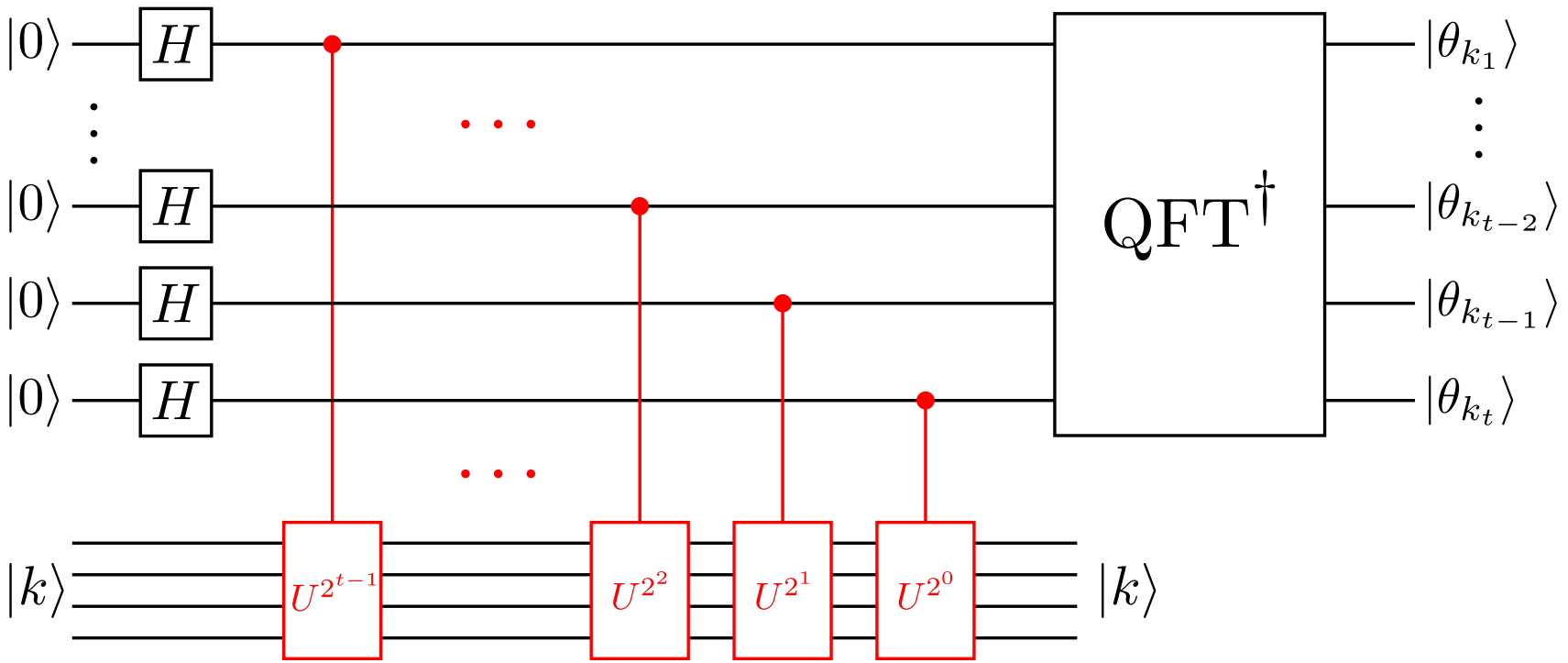
$$\theta_k = 0.\theta_{k_1} \cdots \theta_{k_t}$$



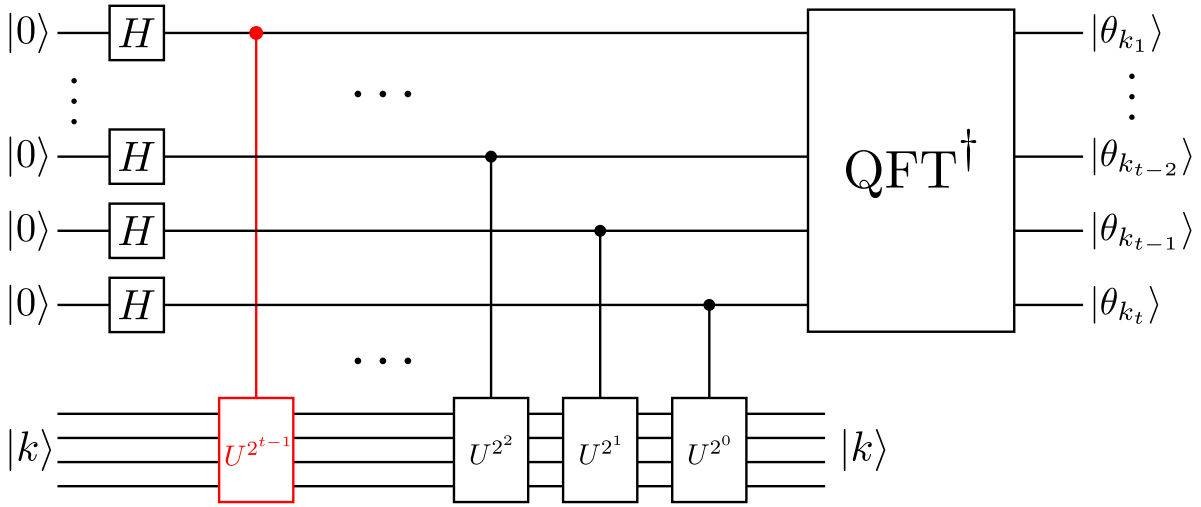
Quantum phase estimation: step 1



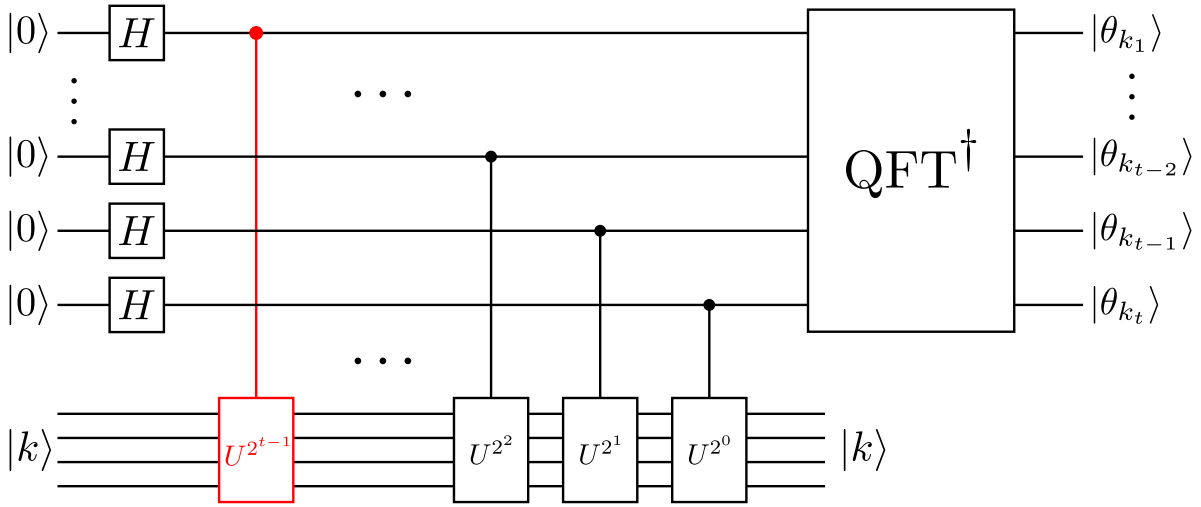
Quantum phase estimation: step 2



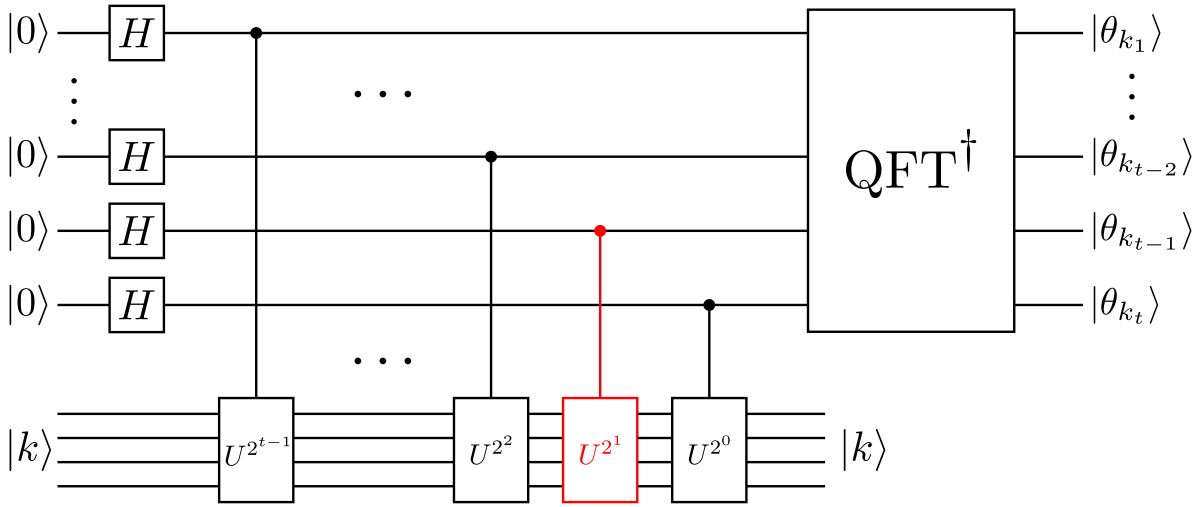
Quantum phase estimation: step 2



Quantum phase estimation: step 2

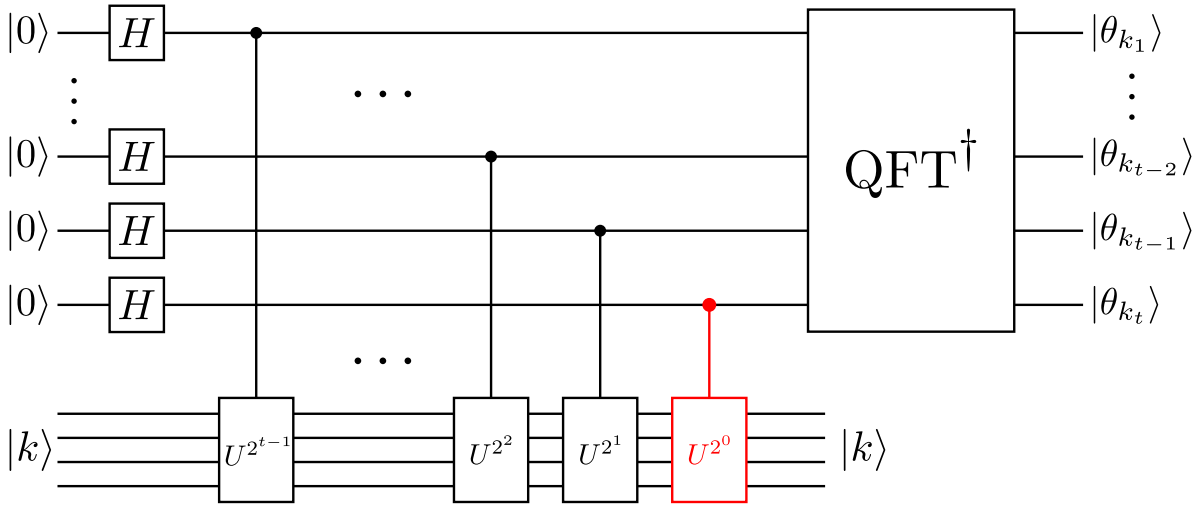


Quantum phase estimation: step 2



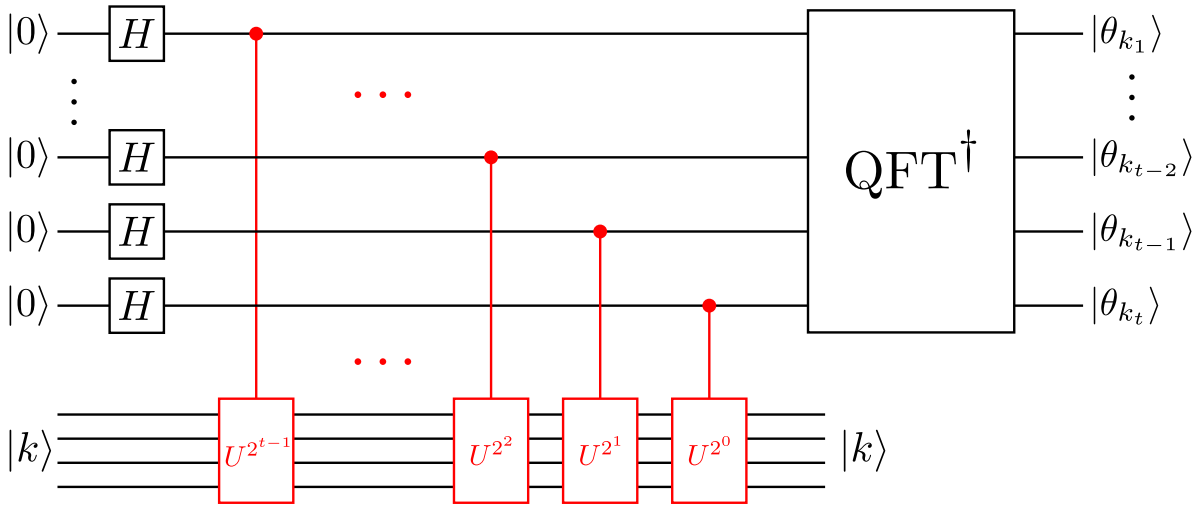
Check second-last qubit (ignore the others)

Quantum phase estimation: step 2



Can show in the same way for the last qubit (ignore others)

Quantum phase estimation: step 2



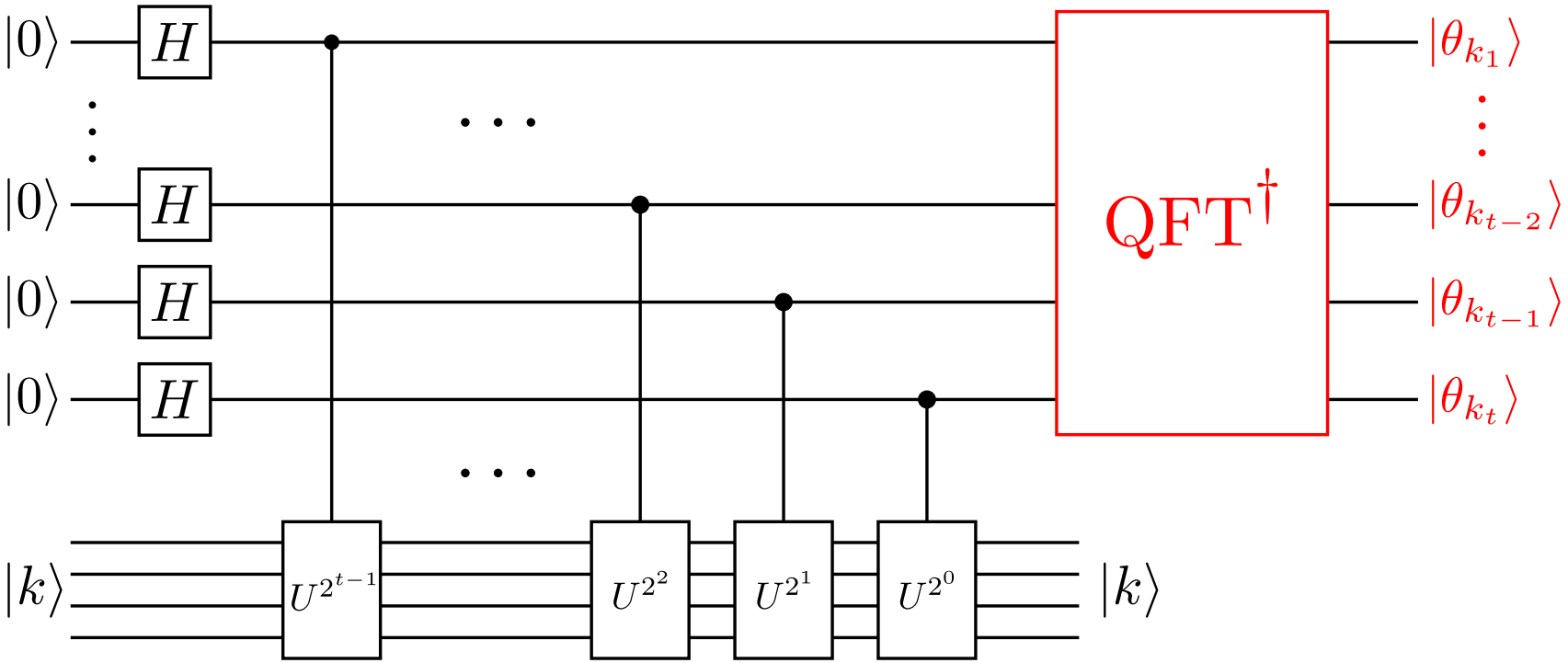
After step 2, we have the state

$$\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.\theta_{k_t}}|1\rangle) \cdots \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.\theta_{k_2}\cdots\theta_{k_t}}|1\rangle)\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.\theta_{k_1}\cdots\theta_{k_t}}|1\rangle)|k\rangle$$

Should look familiar!

Quantum phase estimation: step 3

Measure to learn the bits of θ_k.



Example: QPE for the T gate

Let's apply QPE to estimate the phase of an eigenstate of T: $|1\rangle$.

1. What answer do we expect?
2. How many estimation bits?
3. What does the circuit look like?

Quantum phase estimation

Recall our initial questions... any insights about the second two?

- How to design a circuit that extracts θ_k
- What precision can we estimate it to
- What to do if we don't know $|k\rangle$ in advance

Modular arithmetic

Consider the finite set of integers $\{0, 1, \ldots, N-1\}$.

Basic arithmetic modulo N:

Modular inverses

The inverse of a modulo N (a^{-1}) is an integer b such that

A modular inverse only exists when a and N are *coprime*.

Example: $N = 5$

Example: $N = 4$

Modular exponentiation

Often simplified using techniques such as repeated squaring.

Example: evaluate 8^6 mod 11

Method 1:

Method 2:

Method 3:

Modular arithmetic in PennyLane

PennyLane contains *built-in* operations:

- `qp.Adder`: $|x\rangle \to |x + y \bmod N\rangle$
- `qp.OutAdder`: $|x\rangle|y\rangle|b\rangle \to |x\rangle|y\rangle|b + x + y \bmod N\rangle$
- `qp.Multiplier`: $|x\rangle \to |xy \bmod N\rangle$
- `qp.OutMultiplier`: $|x\rangle|y\rangle|b\rangle \to |x\rangle|y\rangle|b + xy \bmod N\rangle$
- `qp.ModExp`: $|x\rangle|y\rangle \to |x\rangle|y \cdot a^x \bmod N\rangle$

These implementations actually leverage the QFT under the hood; see explanation in this PennyLane demo*.

*https://pennylane.ai/qml/demos/tutorial_shors_algorithm_catalyst#appendix-fun-with-fourier-transforms

Order finding

Input: integers a and N, where $N \geq 2$ and $\gcd(a, N) = 1$
Output: smallest integer r for which $a^r \equiv 1 \bmod N$

Exercise: find the order of $a = 5$ for $N = 7$.

Order finding vs. period finding

Consider the function

Order finding is equivalent to *period finding* for $f_{N,a}(x)$:

Order finding on a quantum computer

Define a unitary operation that performs

If r is the order of a modulo N, and we apply $U_{N,a}$ r times,

Order finding on a quantum computer

$U_{N,a}$ has the same order, r. We can use QPE to efficiently find r with a clever initialization of the target register.

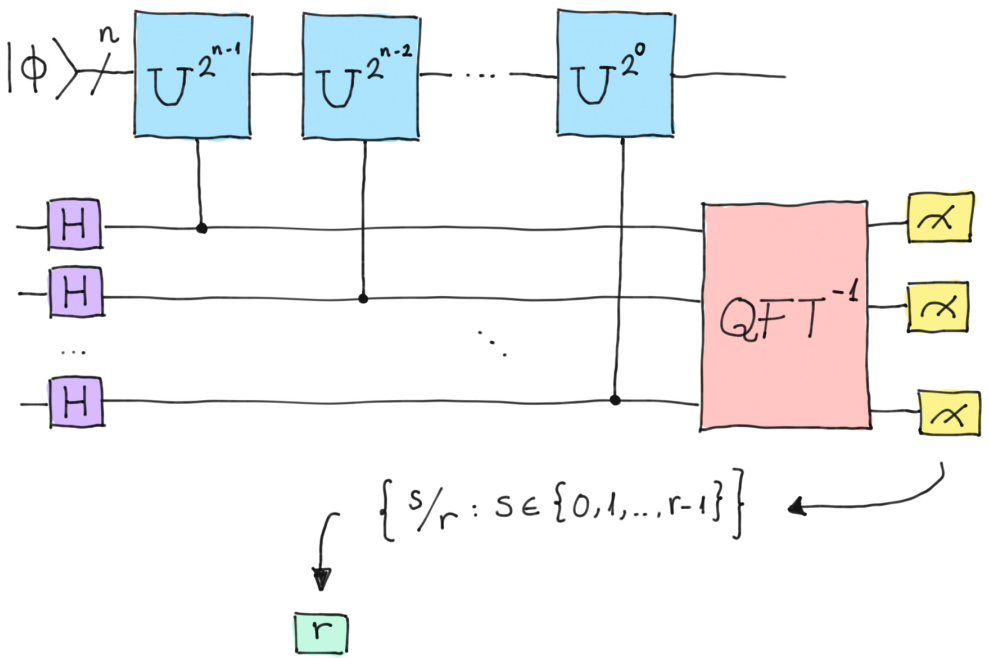


Image credit: Xanadu Quantum Codebook node S.3

Order finding on a quantum computer

Suppse there exists a state $|\phi\rangle$ for which

Consider the state

Apply U to $|\Psi_0\rangle$:

Order finding on a quantum computer

Now consider the state

Apply U to $|\Psi_1\rangle$:

Order finding on a quantum computer

This generalizes to $|\Psi_s\rangle$

It has eigenvalue

Idea: create *any* $|\Psi_s\rangle$, run QPE to get estimate of s/r, compute r.

What's the problem?

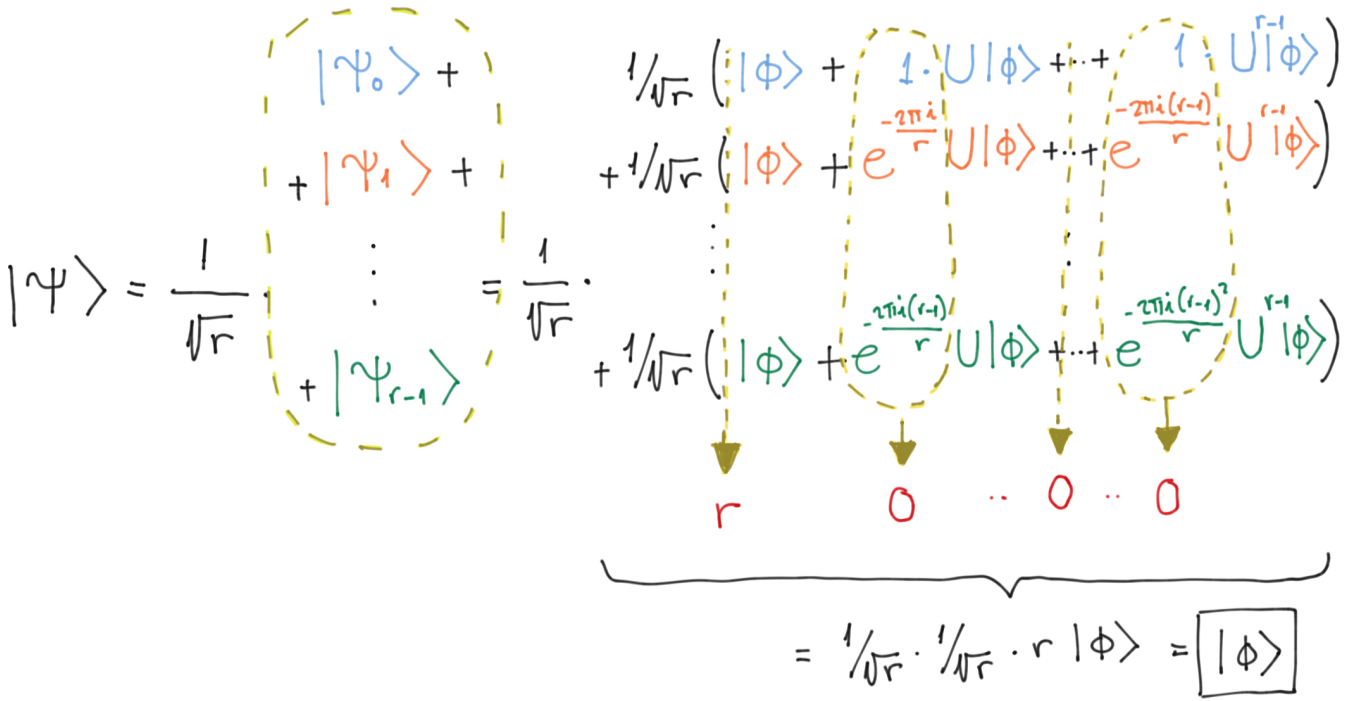
Order finding on a quantum computer

Solution: construct the uniform superposition of all of them.

But what does this equal?

Order finding on a quantum computer

The superposition of all $|\Psi_s\rangle$ is just our original state $|\phi\rangle$!



If we run QPE, the output will be s/r for one of these states.

Image credit: Xanadu Quantum Codebook node S.3

Next time

Content:

- Factoring with Shor's algorithm

Action items:

1. Tutorial Assignment 2 due tomorrow 23:59
2. Midterm checkpoint report due Monday **9:00am**

Recommended reading:

- Codebook modules QFT, QPE, SH
- Nielsen & Chuang 5.3, Appendix A.5

Eigenvalues of unitary matrices

Fun fact: eigenvalues of unitaries are complex numbers with magnitude 1.

Proof: Let λ_k be the eigenvalue associated with eigenvector $|k\rangle$ of a unitary U:

$$U|k\rangle = \lambda_k|k\rangle$$

Take the conjugate transpose:

$$\langle k| U^\dagger = \langle k| \lambda_k^*$$

Multiply the two sides together:

$$\begin{aligned} \langle k| U^\dagger U|k\rangle &= \langle k| \lambda_k^*\lambda_k|k\rangle \\ \langle k|k\rangle &= |\lambda_k|^2\langle k|k\rangle \\ 1 &= |\lambda_k|^2 \end{aligned}$$